

ASSIGNMENT – 3

Advanced Mathematics

B.Sc (Hons.) Agriculture / B.Tech Biotechnology

Beginner Level

Q1. Differentiate:

$$x^3 + 5x^2 + 7$$

Q2. Differentiate:

$$\sin x$$

Q3. Find:

$$\frac{d}{dx}(4x^5)$$

Intermediate Level

Q4. Differentiate using product rule:

$$x^2 \cos x$$

Q5. Differentiate using quotient rule:

$$\frac{x^2 + 1}{x + 1}$$

Q6. Differentiate:

$$(2x^2 + 1)^4$$

Advanced Level

Q7. Find:

$$\frac{d}{dx} \left(\frac{\sin x}{x} \right)$$

Q8. Differentiate implicitly:

$$x^2 + y^2 = 25$$

Q9. Find second derivative:

$$y = x^4 - 3x^2 + 1$$

Beginner Level

Q1. Define matrix.

Q2. Add:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & 1 \\ 5 & 2 \end{bmatrix}$$

Q3. Find:

$$2A$$

Intermediate Level

Q4. Multiply:

$$AB$$

Q5. Find transpose of:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Q6. Find determinant:

$$\begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix}$$

Advanced Level

Q7. Find inverse of:

$$\begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$$

Q8. Solve using matrix method:

$$2x + y = 5$$

$$x - y = 1$$

Q9. Verify:

$$(A + B)^T = A^T + B^T$$

Beginner Level

Q1. Integrate:

$$\int x^2 dx$$

Q2. Integrate:

$$\int \cos x dx$$

Q3. Integrate:

$$\int (3x + 2) dx$$

Intermediate Level

Q4. Integrate:

$$\int x \sin x dx$$

Q5. Evaluate:

$$\int \frac{1}{x} dx$$

Q6. Solve:

$$\int 2x(x^2 + 1)^3 dx$$

Advanced Level

Q7. Evaluate:

$$\int_0^1 (x^2 + 1) dx$$

Q8. Find area under curve:

$$y = x^2$$

between:

$$x = 0 \text{ and } x = 2$$

Q9. Solve:

$$\int \frac{x}{x^2 + 4} dx$$

— END OF ASSIGNMENT 3 —